



**Food and Agriculture
Organization of the
United Nations**

FIAA/C1187 (En)

**FAO
Fisheries and
Aquaculture Circular**

ISSN 2070-6065

SHRIMP INFECTIOUS MYONECROSIS STRATEGY MANUAL



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Bottom left: Transmission electron micrograph of purified infectious myonecrosis virions. Photo credit: ©2006, Microbiology Society, a derivative of figures by Poulos *et al.*, 2006. Journal of General Virology.

Bottom right: Farmers cleaning pond bottom in an Indonesia shrimp farm. Photo credit: ©Mr Hanggono Bambang, Situbondo, Indonesia.

SHRIMP INFECTIOUS MYONECROSIS STRATEGY MANUAL

by

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Under

FAO project TCP/INT/3501

Strengthening biosecurity governance and capacities for dealing with the serious shrimp infectious myonecrosis virus (IMNV) disease

Required citation:

Tang, K.F.J., Bondad-Reantaso, M.G. & Arthur, J.R. 2019. *Shrimp infectious myonecrosis strategy manual*. FAO Fisheries and Aquaculture Circular No. 1187. Rome, FAO.

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ISBN 978-92-5-131798-3

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PREPARATION OF THIS DOCUMENT

This *Shrimp infectious myonecrosis strategy manual* provides guidance in the development of contingency plans for countries, producers and other stakeholders with regard to outbreaks of infectious myonecrosis (IMN), a viral disease of farmed marine penaeid shrimp that is listed by the World Organisation for Animal Health (OIE).

The document is based, in large part, on information presented and discussed by participants at three inter-regional workshops of the FAO project TCP/INT/3501: Strengthening biosecurity governance and capacities for dealing with the serious shrimp infectious myonecrosis virus (IMNV) disease, held in Beijing, People's Republic of China (9–11 November 2016), Natal, Brazil (3–6 May 2017), and Qingdao, People's Republic of China (2–5 November 2017). The workshops were organized by the FAO Aquaculture Branch (FIAA). The first and third workshops were hosted by the Yellow Sea Fisheries Research Institute (YSFRI) and the National Fisheries Technology Extension Center, People's Republic of China. The second workshop was hosted by the Government of Brazil's Ministry of Industry, Foreign Trade and Services (MDIC), Ministry of Agriculture, Livestock and Food Supply (MAPA) and by the Brazilian Association of Shrimp Growers (ABCC). The country delegates and resource experts to this project can be found in Annex 1. The scope of the document follows the section on Disease Strategy Manual, one of a number of Technical Plans of a Contingency Plan from a technical guidance document¹ that FAO produced on *Preparedness and response to aquatic animal health emergencies in Asia: guidelines*.

The project contributes to the FAO Strategic Programme to increase resilience of livelihoods to threats and crises (SP5), particularly two outcomes, namely: Outcome 5.2 - Countries made use of regular information and early warning against potential, known and emerging threats; and Outcome 5.4 - Countries prepared for and managed effective responses to disasters and crises.

¹Arthur, J.R., Baldock, F.C., Subasinghe, R.P. & McGladdery, S.E. 2005. *Preparedness and response to aquatic animal health emergencies in Asia: guidelines*. FAO Fisheries Technical Paper No. 486. Rome, FAO. 40 pp.

ABSTRACT

This *Shrimp infectious myonecrosis strategy manual* provides key information for national policy-makers relevant to the development of contingency plans for countries, producers and other stakeholders with regard to outbreaks of infectious myonecrosis (IMN), a viral disease of farmed marine penaeid shrimp that is listed by the World Organisation for Animal Health (OIE). IMN is a viral disease, discovered in 2002, that has caused substantial mortalities in populations of cultured Pacific whiteleg shrimp (*Penaeus vannamei*) initially reported in Brazil (2002) and Indonesia (2006) and recently in India (2016) and Malaysia (2018). This virus infects, and causes focal to extensive white, opaque necrotic lesions in, the skeletal muscle. Infection can cause 40 to 70 percent mortality of populations in affected ponds, and economic losses due to this disease during 2002–2011 amounted to over USD 1 billion. Further outbreaks of IMN, particularly in areas that are currently free of this disease, would be expected to have similar devastating effects on local shrimp producers and the surrounding communities; and thus, there is an urgent need to develop contingency plans to control, contain and eradicate the IMN virus (IMNV) that causes this disease.

The purpose of this manual is to provide support for the various components of a national contingency plan. The information provided includes: (1) the nature of IMN: providing a brief review of disease etiology, susceptible species and global distribution; (2) diagnosis of infection: describing the gross clinical signs of disease, field diagnostic methods, differential and laboratory methods for diagnosis; (3) prevention and treatment: providing information on vaccination, and resistance and immunity of the hosts; (4) epidemiology: providing information on IMNV's geographic distribution, persistence in the environment, modes of transmission, vectors and reservoir hosts, factors influencing disease transmission and expression, and impact of the disease; (5) principles of control and eradication: describing the methods for control, containment and eradication of IMNV, and trade and industry considerations; and (6) policy development and implementation: summarizing the overall policy, IMN-specific objectives, problems, overview of response options, strategies for eradication and control, capacity building and funding and compensation.

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ACKNOWLEDGEMENTS

This publication was supported by the FAO project TCP/INT/3501: Strengthening biosecurity governance and capacities for dealing with the serious shrimp infectious myonecrosis virus (IMNV) disease that was participated by the following countries: Brazil, Ecuador, Indonesia, Mexico, People's Republic of China and Thailand.

The Yellow Sea Fisheries Research Institute (YSFRI) and the National Fisheries Technology Extension Center, People's Republic of China, the Government of Brazil, Ministry of Industry, Foreign Trade and Services (MDIC), Ministry of Agriculture, Livestock and Food Supply (MAPA), and the Brazilian Association of Shrimp Growers (ABCC) are gratefully acknowledged for graciously hosting the workshops. We thank the contributions of all workshop participants to this effort.

In addition, we would like to thank Dr David Cummins of the Commonwealth Scientific and Industrial Research Organization (CSIRO), Australia; Ms Mukti Sri Hastuti of the Ministry of Marine Affairs and Fisheries of Indonesia; Dr Christa L. Speekmann of the United States Department of Agriculture-Animal and Plant Health Inspection Service; Mr Marcelo Lima Santos of the Brazilian Association of Shrimp Growers (ABCC), Brazil; Dr Passos de Andrade Thales of the Universidade Estadual do Maranhao (UEMA), Brazil and Mr Hanggono Bambang of the Brackishwater Aquaculture Development Center-Situbondo, Indonesia for taking their time to review this document and provide helpful comments.

The officials and staff of the FAO representation offices in the People's Republic of China and Brazil, and the Aquaculture Branch (FIAA – O. Elhassan, L. Falcone, E. Irde) and the Statistics and Information Branch (FIAS – M. Guyonnet) of the FAO Department of Fisheries and Aquaculture are also gratefully acknowledged for operational and logistical support during the preparation and implementation of the project and finalization of this document.

ABBREVIATIONS AND ACRONYMS

aa	amino acids
AAH	Aquatic animal health
ABCC	Associação Brasileira de Criadores de Camarão (Brazilian Association of Shrimp Growers)
AFA	Alcohol-formalin-acetic acid
BMPs	Better management practices
bp	base pairs
DSRM	Double-stranded RNA-binding motif
dsRNA	Double-stranded RNA
FAO	Food and Agriculture Organization of the United Nations
FCR	Feed conversion ratio
FIAA	The FAO Aquaculture Branch
FIAS	The FAO Fisheries and Aquaculture Statistics and Information Branch
H&E	Haematoxylin and eosin
ICT	Information and communication technology
IHC	Immunohistochemistry
IHHNV	Infectious hypodermal and haematopoietic necrosis virus
iiPCR	Insulated isothermal polymerase chain reaction
IMN	Infectious myonecrosis
IMNV	Infectious myonecrosis virus
ISH	<i>in situ</i> hybridization
kDa	kilodaltons
LFD	Lateral flow dipstick
MAPA	Ministry of Agriculture, Livestock and Food Supply
MCP	Major capsid protein
MDIC	Ministério do Desenvolvimento, Indústria, e Comércio Exterior (Ministry of Industry, Foreign Trade and Services)
MrNV	<i>Macrobrachium rosenbergii</i> nodavirus
NBT/BCIP	Nitro blue tetrazolium/5-bromo-4-chloro-3-indolyl phosphate
NSAAH	National strategy for aquatic animal health
nt	Nucleotide
OIE	World Organisation for Animal Health
ORFs	Open reading frames
PL	Postlarvae
PMP	Progressive management pathway
PvNV	<i>Penaeus vannamei</i> nodavirus
RdRp	RNA-dependent RNA polymerase
RNA	Ribonucleic acid
RNAi	RNA interference
RT-LAMP	Reverse transcription loop-mediated isothermal amplification
RT-PCR	Reverse transcription polymerase chain reaction
RT-qPCR	Quantitative reverse transcription polymerase chain reaction
SPF	Specific-pathogen-free
SPR	Specific-pathogen-resistant

SWOT	Strengths weaknesses opportunities and threats
TEM	Transmission electron microscopy
TSV	Taura syndrome virus
UTR	Untranslated region
WGS	Whole genome sequencing
WSSV	White spot syndrome virus

1. INTRODUCTION

Aquaculture is the most rapidly growing area of the global agriculture sector. The most important aquaculture production sector from an economic standpoint is the farming of marine shrimp. The annual value of production of marine shrimp is estimated at USD 40 billion on the world market (FAO, 2016). During the past three decades, viral and bacterial diseases have emerged as serious impediments to shrimp farming, and many of these pathogens have spread around the globe. The resulting pandemics have collectively cost the global shrimp aquaculture industry billions of dollars. Of the over 30 microorganisms now known to infect penaeid shrimp, eight viral and bacterial diseases dominate the list in their adverse effects and have been listed as diseases notifiable to the World Organisation for Animal Health (OIE, 2018a). Responses of the aquaculture industry to the threats of aquatic diseases include the development of country-specific contingency plans for specific diseases (Arthur *et al.*, 2005).

Among these OIE-notifiable diseases is infectious myonecrosis (IMN), which is caused by the infectious myonecrosis virus (IMNV). IMNV infects penaeid shrimp, including the most commonly farmed species, the whiteleg shrimp (*Penaeus vannamei*) and the giant tiger prawn (*P. monodon*). This disease is important, as the mortalities in IMNV-infected populations can reach to 70 percent. Losses due to IMNV from 2002 through 2011 were estimated to be more than USD 1 billion (Lightner *et al.*, 2012). IMNV was discovered in farmed shrimp in Brazil in 2002 and subsequent outbreaks occurred in Indonesia, reported initially in 2006, in India, reported in 2016 and most recently in Malaysia in 2018. On a global scale, these countries produce approximately 27 percent of farmed shrimp, so strategies to prevent catastrophic losses to the industry are important. This manual provides information specific to IMN relevant to the development of such strategies, reviews the existing knowledge for this disease and provides recommendations for its prevention and control, including detailed measures for containment and eradication. The manual can serve as a framework for the development of a national contingency plan to IMNV outbreaks.

2. THE NATURE OF INFECTIOUS MYONECROSIS

IMN is a viral disease that causes substantial mortalities in farmed populations of *P. vannamei*. The principal target tissue for IMNV infection is the striated muscle (skeletal and less often, cardiac muscle). IMNV-infected shrimp present focal to extensive white necrotic areas in striated muscle, especially in the distal abdominal segments and tail fan. The causative agent was identified as a virus (IMNV) in 2003. The disease has caused severe losses to shrimp producers in both Brazil and Indonesia (Figure 1), and in India and Malaysia. Thus, IMNV was listed by the OIE in 2005 (OIE, 2018a).

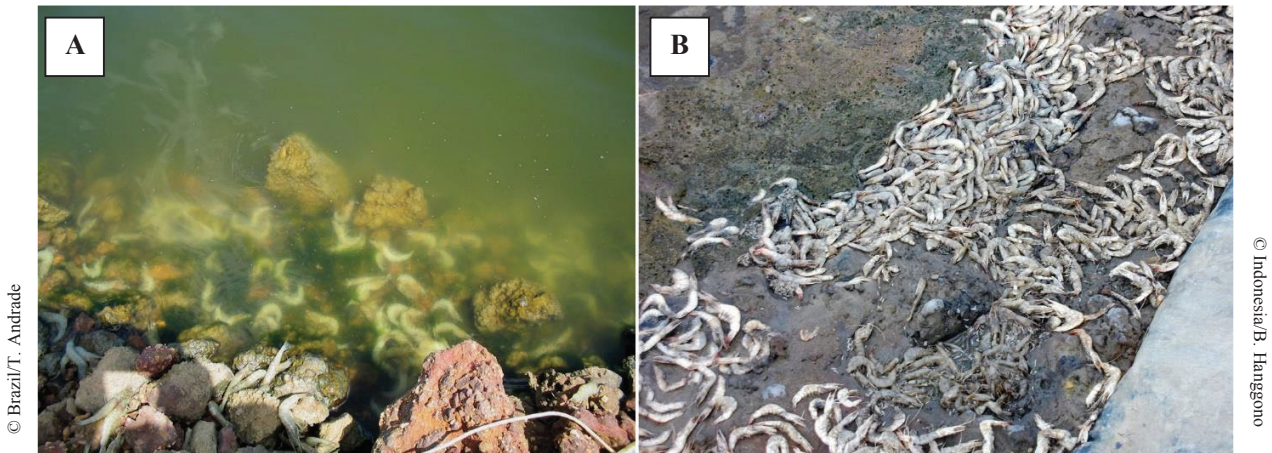


Figure 1. Mass mortality of cultured *Penaeus vannamei* caused by infectious myonecrosis in farms located in (A) northeastern Brazil in 2002; (B) East Java, Indonesia in 2006.

2.1 Etiology

IMNV is classified as a member of the family *Totiviridae* and has been found to be most closely related to *Giardia lamblia* virus (Poulos *et al.*, 2006; Nibert, 2007), a virus infecting a flagellated parasite of mammals and birds.

IMNV virions are icosahedral in shape and 40 nm in diameter, with a buoyant density of 1.366 g/ml in cesium chloride (CsCl) (Figure 2). The genome consists of a single, double-stranded ribonucleic acid (dsRNA) molecule of 8226–8230 base pairs (bp) (Loy *et al.*, 2015; Naim *et al.*, 2015). Sequencing of the viral genome reveals two non-overlapping open reading frames (ORFs) (Figure 3). The first ORF (ORF1, nucleotide, nt, 470–5596) encodes a polyprotein in the range of 1653–1708 amino acids (aa) that is processed co- or post-translationally into four peptides, designated 1–4 (Nibert, 2007). The peptide 1 (in the range of 141–196 aa) contains a dsRNA-binding motif (DSRM); this peptide is thought to be involved in blocking host antiviral responses (Naim *et al.*, 2015). The peptide 2 (284 aa, 31 kilodaltons, kDa) and/or peptide 3 (327 aa, 36 kDa) is thought to form the fibers protruding from the IMNV virions (Tang *et al.*, 2008). The peptide 4 is a major capsid protein (901 aa), as determined by amino acid sequencing, with a molecular mass of 106 kDa (Poulos *et al.*, 2006). The second ORF (ORF2, nt 5884–8133), in the -1 frame relative to ORF1, encodes the putative RNA-dependent RNA polymerase (RdRp, 736 aa) with a molecular mass of 85 kDa.

Through cryomicroscopy and 3D imaging reconstruction using the purified IMN virions, there are protrusion fibers that form five-fold axes; these protrusions may play a role in facilitating extracellular transmission and pathogenesis of IMNV (Tang *et al.*, 2008). In addition, from a yeast two-hybrid assay, the laminin receptor is found to interact with the IMNV capsid protein, suggesting that the laminin receptor may be the receptor for IMNV (Busayarat *et al.*, 2011).

The complete genomes of IMNV types originating from Brazil and Indonesia have been sequenced and found to be 99.6 percent identical at the nucleotide level (Poulos *et al.*, 2006; Senapin *et al.*, 2007). The 99.6 percent full genome sequence identity (and anecdotal information on the transfer of stocks of *P. vannamei*) indicates that the disease was introduced from Brazil to Indonesia in 2006.

IMNV sequences from both Brazil and Indonesia were analyzed, and its nucleotide substitution rates was estimated to be approximately 1×10^{-3} substitutions/site/year (Naim, Brown and Nibert, 2014). This evolutionary rate is at the high end of dsRNA (double-stranded RNA) viruses (Stenger, Sisterson and French, 2010), indicating that IMNV is a rapidly evolving dsRNA virus.

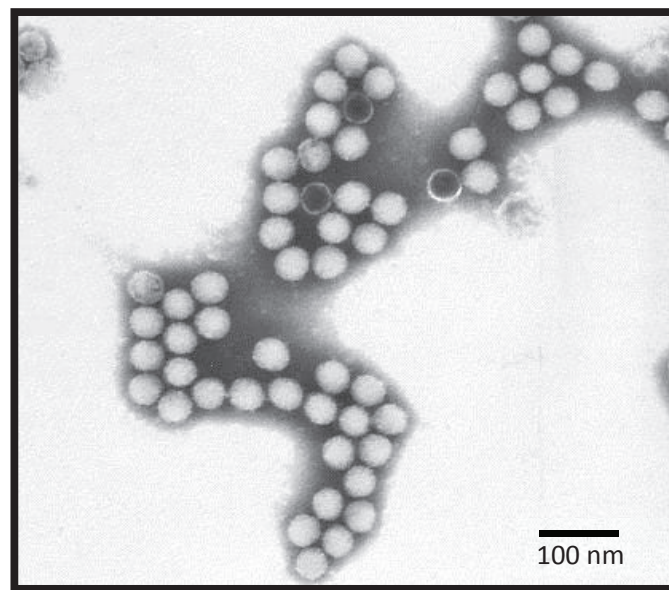


Figure 2. Transmission electron micrograph of infectious myonecrosis virus (IMNV). Caesium chloride-gradient fraction of purified virions. Stained with 2% phosphotungstic acid.

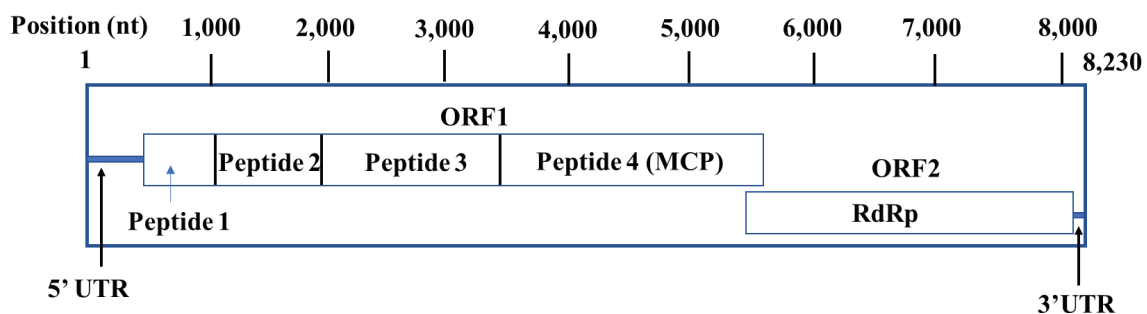


Figure 3. Infectious myonecrosis virus (IMNV) genome organization: 5' and 3' UTRs (untranslated regions), ORFs, major capsid protein (MCP) and RNA-dependent RNA polymerase (RdRp).

2.2 Susceptible species

Species that fulfill the criteria for listing as susceptible to infection with IMNV according to chapter 1.5 of the OIE's *Aquatic Animal Health Code* (the "Aquatic code", OIE 2018a) include brown tiger prawn (*Penaeus esculentus*), banana prawn (*P. merguensis*), whiteleg shrimp (*P. vannamei*) and giant tiger prawn (*P. monodon*).

Species for which there is incomplete evidence for susceptibility according to chapter 1.5 of the Aquatic code include blue shrimp (*P. stylirostris*) and southern brown shrimp (*P. subtilis*). *Penaeus stylirostris* has been experimentally infected with IMNV, but mortalities did not occur in the laboratory infection trials (Tang *et al.*, 2005). *Penaeus subtilis* is found in the offshore waters of Brazil and has long been considered a potential species for aquaculture (Nunes, Gesteira and Goddard, 1997). Individuals of this species were used in laboratory challenge studies where it was shown by IMNV reverse transcription polymerase chain reaction (RT-PCR) that they can be infected with IMNV resulting in a subsequent mortality of approximately 5 percent (Coelho *et al.*, 2009).

2.3 Global distribution

IMN was first reported in populations of cultured *P. vannamei* at a shrimp farm in the State of Piauí, Brazil in September 2002 (Nunes, Cunha-Martins and Vasconcelos-Gesteira, 2004). IMNV was first detected in Indonesia in early 2006 (Senapin *et al.*, 2007) and has been reported from the islands of Java (East Java and West Java), Sumatra, Bangka, Borneo (West Borneo), Sulawesi (South Sulawesi), Bali, Lombok and Sumbawa (Senapin *et al.*, 2007). IMNV was detected in West Bengal State of India in July 2016 (Sahul Hameed *et al.*, 2017), and spread to Andhra Pradesh State, the largest farming region for *P. vannamei* in India, in April 2017. The most recent case occurred in the State of Melaka, Malaysia in June 2018 (OIE, 2018c).

3. DIAGNOSIS OF INFECTION

Shrimp affected by IMN exhibit distinctive, but not specific, gross clinical signs. Examination of histological sections of skeletal muscle, the primary targeted tissue, stained with haematoxylin and eosin (H&E) can provide a presumptive diagnosis of IMNV infection. However, for definitive diagnosis, and to determine the status of clinically normal shrimp, RT-PCR or quantitative reverse transcription polymerase chain reaction (RT-qPCR) testing is recommended (OIE, 2018b). Clinical chemistry methods, such as *in situ* hybridization (ISH) and immunohistochemistry (IHC) are not used routinely by shrimp pathologists; also, these methods are not as sensitive as RT-PCR (or RT-qPCR). There is no continuous cell culture system for shrimp; therefore, the diagnosis depends on the gross signs, histopathology, RT-PCR-based methods and immunoassays mentioned above. A flow chart of the diagnosis of IMNV is given in Figure 4.

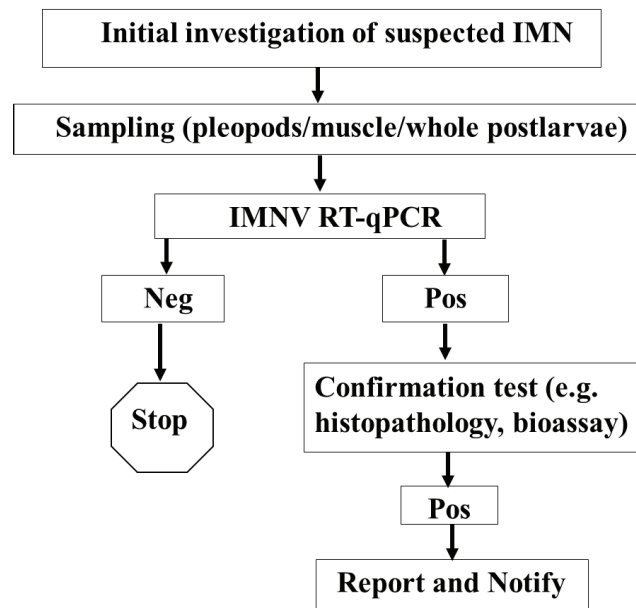


Figure 4. Diagnostic flowchart. Pos: positive, Neg: negative.

In this document, the diagnostic methods make reference to the three levels of diagnosis, i.e. Level I, Level II and Level III as described by Bondad-Reantaso *et al.* (2001).

3.1 Gross clinical signs (Level I)

Shrimp in the acute phase of IMN present focal to extensive white necrotic areas in striated (skeletal) muscles (Figure 5A), especially of the distal abdominal segments and tail fan, that become reddened after death, similar to the colour of cooked shrimp (Figure 5B) (Poulos *et al.*, 2006). The distinctive clinical sign, whitish, opaque, muscle, is usually more prominent in *P. vannamei* than in either *P. stylirostris* or *P. monodon*. In addition, the presence of these clinical conditions is not specific to IMN and therefore, cannot be used for definitive diagnosis.

IMN can occur at any stage of shrimp grow-out and is a chronic infection. The mortality can reach to 70 percent at the end of the culture cycle (Nunes, Cunha-Martins and Vasconcelos-Gesteira, 2004). Evidence of the disease, other than the appearance of whitish tail muscle, is that the shrimp become lethargic and have reduced feed consumption.



Figure 5. Gross signs of infectious myonecrosis. (A) Farmed *Penaeus vannamei* from an outbreak in Brazil, exhibiting various degrees of skeletal muscle necrosis visible as an opaque, whitish discoloration in the abdomen; (B) farmed *P. vannamei* from an outbreak in Indonesia, showing reddened necrotic tails.

3.2 Field diagnostic methods

The diagnosis of IMNV in shrimp is usually performed via a sequence of procedures after a preliminary diagnosis based on gross clinical signs. These include histological demonstration of lesions, which is usually too slow to allow practical management decisions by shrimp farmers. Molecular diagnostic methods are much faster and can provide immediate information to shrimp farmers. The molecular detection of IMNV has been successfully employed, with RT-PCR (and RT-qPCR) being widely used for the rapid and specific detection of the virus (OIE, 2018b).

3.2.1 Rapid on-site diagnostic assays (Level III)

Optimal on-site detection systems should be rapid, sensitive, inexpensive and easy to maintain and operate by non-specialists. In addition, the reagents should be provided in a format that allows easy shipping and storage. There are a range of different technologies currently available for the rapid on-site diagnosis of IMNV:

- Reverse transcription loop-mediated isothermal amplification (RT-LAMP) combined with a lateral flow dipstick (LFD) has been developed; the total assay interval is less than 75 min (Andrade, 2007; Andrade and Lightner, 2009; Puthawibool *et al.*, 2009). These assays are carried out with a simple water bath and are inexpensive.
- A strip test (lateral flow immunochromatographic assay, LFIA) using two monoclonal antibodies to detect the IMNV MCP has been developed by Chaivisuthangkura *et al.* (2013). This immunoassay is not as sensitive as PCR-based methods, but it has the advantages of lower-cost and speed (it gives results within 15 min after sample preparation), and is simple in format, not requiring expensive equipment or specialized skills. Thus, it is appropriate for use by farmers for confirming the presence of IMNV.

- A commercial IMNV detection assay, based on insulated isothermal polymerase chain reaction (iiPCR, based on the RT-qPCR principle) is also available for on-site detection of IMNV. The analyzer is called POCKIT™ (manufactured by GeneReach Biotechnology Corp.) and is certified by OIE. The assay can be completed within 60 min. A hand-held device (POCKIT™ Micro) is also available for pond-site detection of IMNV.

These types of rapid diagnostic tests need to be evaluated by local field testings, after which their use can then be incorporated into national contingency plans.

3.3 Differential diagnosis

When whitish, necrotic tail muscle is observed in dead and moribund shrimp, especially in samples from Brazil, Indonesia, India and Malaysia, IMNV should be included in the list of diseases for diagnosis. However, this clinical sign can also be caused by a number of factors other than IMNV infection, e.g. hypoxia, crowding, sudden changes in temperature or salinity (Lightner, 1988).

IMN is different from whitetail diseases found in penaeid shrimp or in the giant river prawn (*Macrobrachium rosenbergii*) (family Palemonidae), although the clinical and histological signs are similar. These non-IMNV whitetail diseases exhibit gross and histological signs that mimic IMN, but are caused by nodaviruses: *Penaeus vannamei* nodavirus (PvNV) (Tang *et al.*, 2007) and *Macrobrachium rosenbergii* nodavirus (MrNV) (see chapter 2.2.6 of the *Manual of Diagnostic Tests for Aquatic Animals* (the "Aquatic manual", OIE, 2018b)). To differentiate between IMNV and PvNV in *P. vannamei*, and MrNV in *M. rosenbergii*, RT-PCR (RT-qPCR) with primers specific to each virus and/or ISH with specific probes/or virus-specific immunodetection assays should be used.

Specific molecular tests such as RT-PCR (RT-qPCR), ISH or immunodetection assays can be used to differentiate IMNV infection from other, non-viral causes. IMNV-associated mortality is chronic, progressing relatively slow. However, this type of mortality can also be caused by stress-related bacterial infections. Histological examinations and/or bacterial isolation and enumeration can be used to determine such occurrences and differentiate them from IMNV infection.

3.4 Laboratory methods

3.4.1 Sample submission

In the first instance, clinical specimens should be sent to local (regional or state) aquatic animal diagnostic laboratories. If IMNV infection is diagnosed, the Competent Authority should be notified, and the diagnosis confirmed through repeated testing with other methods and/or testing of additional specimens from the suspected IMNV-infected populations. The Competent Authority should be contacted directly to obtain information on what additional clinical materials may be required and how they should be collected, stored and transported to satisfy requirements for confirming the original diagnosis of IMNV infection.

The number of individual samples required when screening shrimp for IMNV will depend on the population size and level of confidence at which infection needs to be excluded (Lightner, 1996). Ideally, all laboratory procedures should comply with the Aquatic manual (OIE, 2018b). The recommended sample size (i.e. number of individual shrimp) that should be collected for diagnosis when IMNV is present in the population at a prevalence of 2 percent is 149 (Aquatic code, chapter 1.4, OIE, 2018a). This is assuming that the sensitivity and specificity of the method used for diagnosis are both 100 percent. The individual shrimp can be pooled for diagnosis.

For RT-PCR (or RT-qPCR) detection, 50 postlarvae (PL) or 5–10 juvenile shrimp can be pooled into one sample for RNA extraction. The quality of the specimens is important, and they must be properly preserved, stored and transported to avoid RNA degradation. For histological evaluation, the shrimp should be fixed in Davidson's AFA (alcohol-formalin-acetic acid) for 24–48 hr (depending on the shrimp size) and then transferred to 70 percent ethanol for storage. The prolonged storage of infected shrimp in the acidic (pH 3–4) Davidson's fixative will damage the IMNV RNA genome; this may lead to a weak or no reaction after ISH. It is best to split a sample into a Davidson's-fixed sample and a 95 percent ethanol-preserved or frozen sample. Fixed tissues can be submitted for histopathological examination; the frozen or ethanol-preserved tissues can be used for RT-PCR (or RT-qPCR) analyses.

When bioassays are used to confirm the presence of infectious IMNV in the suspected or positive samples, a highly susceptible host species such as *P. vannamei* should be used and then exposed to the IMNV-suspect fresh or frozen (stored at ultra-low temperatures) tissues.

3.4.2 Wet mounts (Level I)

The skeletal muscle can be sampled by biopsy and placed on a microscope slide. A drop of saline (2 percent NaCl) should be added to the slide to keep the tissue from drying out. The slide should be viewed immediately using a phase-contrast microscope by starting at lower magnification (total magnification of 100x) and finishing using higher magnification (400 to 600x). To obtain an accurate diagnosis, the taking of multiple samples is preferred. The IMNV-infected shrimp may show losses of normal striations and/or fragmentation of the muscle fibers. The lymphoid organ can be sampled for wet-mount examinations; in the severely infected shrimp, the lymphoid organs are inflamed and appear whitish in colour (M.L. Santos, personal communication, March 14, 2018).

3.4.3 Histopathology (Level II)

For histopathology, moribund whole shrimp or samples of their muscle tissue can be used. The samples should be preserved in the Davidson's AFA or a RNA-friendly fixative (pH 6–7; this is to reduce RNA degradations for ISH analyses), processed into paraffin blocks, and sectioned. The tissue sections should then be stained with H&E through the use of a standard protocol (Lightner, 1996). Stained sections are then examined using a compound microscope.

Infected tissues exhibit lesions formed by coagulative muscle necrosis and haemocytic infiltration (Figure 6A). Basophilic inclusion bodies are seen within the cytoplasm of skeletal muscle cells (Figure 6B), lymphoid organ, hindgut, and phagocytic cells within the hepatopancreas and heart. In addition, lymphoid organ spheroids are typically seen within the lymphoid organ (Figure 6C); some are ectopic in the heart, antennal gland and loose connective tissues (Figure 6D).

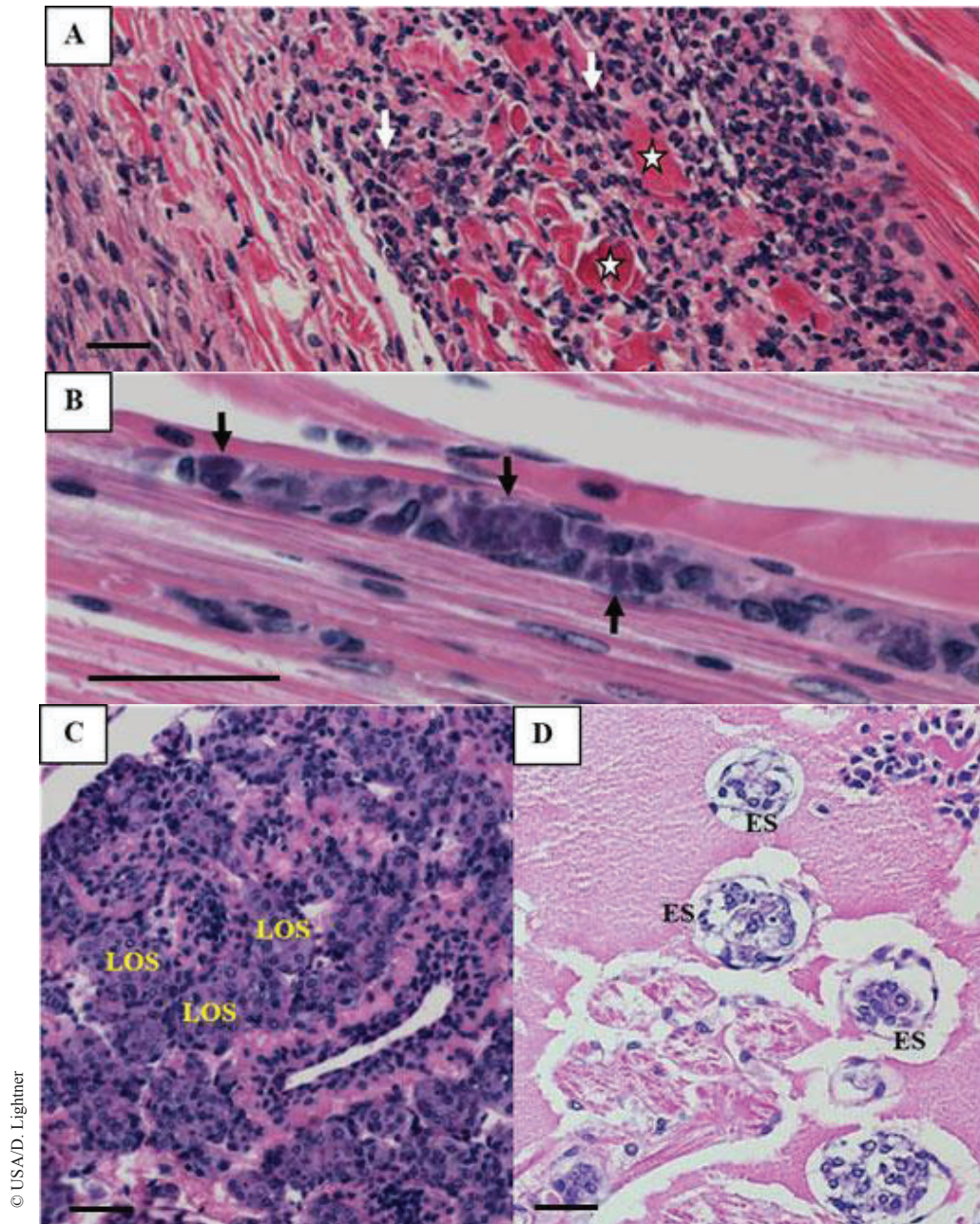


Figure 6. Histopathological characterization of infectious myonecrosis virus (IMNV) infected *Penaeus vannamei*, H&E stain. (A) Coagulative necrosis (stars) of skeletal muscle accompanied by haemocytic infiltration (white arrows); (B) perinuclear basophilic inclusion bodies (black arrows) are evident in the muscle cells; (C) presence of lymphoid organ spheroids; and (D) ectopic spheroids in the heart. Scale bars = 25 µm.

Although histopathology can provide a presumptive diagnosis of IMNV infection, transmission electron microscopy (TEM) or specific molecular tests such as RT-PCR (RT-qPCR), or ISH or immunohistochemistry (IHC) or western blot analyses that detect viral proteins, are required for confirmation (OIE, 2018b). A disease of muscle necrosis associated with low mortalities was reported in *P. vannamei* cultured in Ecuador (Melena *et al.*, 2012). These shrimp exhibited gross signs and histological lesions indistinguishable to those seen in IMN, but IMNV (nor PvNV) was not the pathogen as determined by RT-PCR analyses.

3.4.4 Transmission electron microscopy (Level III)

For TEM examination, specimens with histological lesions indicating the presence of IMNV are preferred. Tissues most suitable for detecting IMN virions by TEM include tail muscle and pleopods. IMN virions are icosahedral in shape and 40 nm in diameter (see Figure 2). Detailed procedures for TEM are available in Lightner (1996).

3.4.5 Molecular techniques (Level III)

(a) RT-PCR

Several RT-PCR and RT-qPCR protocols have been described for the specific and sensitive detection of IMNV in samples. Details on how to perform this test can be found in the original publications (Poulos and Lightner, 2006; Andrade *et al.*, 2007; Senapin *et al.*, 2007) and in the Aquatic manual (OIE, 2018b). Since IMNV genome is dsRNA, the heat denaturation step (95–99 °C for 5–10 min) must be performed prior to adding into the RT-PCR (RT-qPCR) mixture. For confirmation, the amplified products can be sequenced and analyzed with blast searches. Commercial RT-PCR and RT-qPCR kits for detection of IMNV RNA are also available, for example, IQ plus IMNV kit and IQ Real IMNV kit (GeneReach Biotechnology Corp.).

RT-PCR data should be interpreted with caution, particularly when shrimp without clinical signs are tested. Samples with infections not manifesting clinical signs can have IMNV RNA levels approaching the detection limit of the tests. This may lead to a high variation in the test results. A laboratory bioassay can be performed to confirm the presence of IMNV by propagating the viruses to a higher level to be readily detected. In most of cases, RT-PCR should be used in conjunction with histopathology for IMNV detection to corroborate with the IMN diagnostic problems.

(b) *In situ* hybridization (ISH)

Procedures used for ISH were described by Lightner (1996). The fixed shrimp are processed, embedded in paraffin and sectioned. The sections on slides are overlaid with hybridization buffer containing an IMNV probe usually labeled with digoxigenin and placed on a heated surface at 84–90 °C for 10 min, chilled on the ice for 1–2 min and hybridized overnight at 42 °C, followed by an antibody (against digoxigenin) reaction and a final NBT/BCIP (nitro blue tetrazolium/5-bromo-4-chloro-3-indolyl phosphate) colour development (Tang *et al.*, 2005).

(c) Immunodetection

Monoclonal antibodies have been produced to detect IMNV MCP through IHC detection. An anti-IMNV monoclonal antibody was used to detect viral inclusions in the histological tissue sections (Seibert *et al.*, 2010; Kunanopparat *et al.*, 2011). IMNV can be detected in a dot-blot format, applying a combination of three monoclonal antibodies against the tissue homogenate prepared from infected shrimp tissues. This resulted in a two-fold increase in sensitivity over that obtained with the use of a single monoclonal antibody, but the sensitivity was still approximately ten times lower than that of one-step RT-PCR with the same sample (Kunanopparat *et al.*, 2011). From these monoclonal antibodies, a lateral flow immunochromatographic test was also developed for pond-site diagnostics, but the sensitivity is 300 times less than the one-step RT-PCR (Chaivisuthangkura *et al.*, 2013).

3.4.6 Bioassays

Bioassays with susceptible species can be used to determine the presence of infectious viruses in clinical samples. Bioassay protocols that involve injecting purified IMNV virions or tissue homogenate prepared from IMNV-suspect individuals into healthy indicator shrimp have been published (Tang *et al.*, 2005; Poulos *et al.*, 2006). Bioassays can also be performed by feeding IMNV-infected shrimp tissues to indicator shrimp, generally following the procedures described by Lightner (1996). The bioassays can be performed at an elevated temperature (>30 °C), especially when the feeding tissues (or homogenates) contain lower viral loads. The infection of IMNV in the indicator shrimp is then monitored by histopathology, ISH (or IHC) and/or RT-PCR (RT-qPCR).

The methods currently available for surveillance and diagnosis of infection with IMNV are listed in Table 1.

Table 1. Comparison of the suitability of the different methods for surveillance and diagnosis of infection with infectious myonecrosis virus (IMNV)

Method	Targeted surveillance				Presumptive diagnosis	Confirmatory diagnosis
	Larvae	Postlarvae	Juveniles	Adults		
Gross signs	d ¹	d	c	c	c	d
Bioassay	d	d	c	c	c	a
Wet mounts	d	d	c	c	c	c
Histopathology	d	d	b	b	a	c
TEM	d	d	d	d	d	d
Antibody-based assays	d	d	d	d	c	c
ISH	d	d	b	b	a	a
Nested RT-PCR or RT-qPCR, or RT-LAMP	a	a	a	a	a	a
Amplicons sequencing	d	d	d	d	a	a

¹ a = the method is the recommended method for reasons of availability, utility, and diagnostic specificity and sensitivity; b = the method is a standard method with good diagnostic sensitivity and specificity; c = the method has application in some situations, but cost, accuracy or other factors severely limits its application; d = the method is presently not recommended for this purpose (Source: OIE, 2018b).

3.4.7 Confirmation of infection

To confirm cases of IMNV infection, where histopathological characteristics of IMN are present, IMNV RNA or proteins (e.g. MCP) in histological tissue sections can be detected through the use of ISH and/or IHC techniques. When fresh clinical materials are available, infection can also be confirmed with RT-PCR (RT-qPCR) or bioassays in conjunction with the diagnostic methods for IMNV. Of the diagnostic methods available, RT-qPCR is preferred to confirm infection because of its sensitivity, specificity and speed.

The advantages and disadvantages associated with commonly used laboratory tests for diagnosing IMNV infection are summarized in Table 2.

Table 2. Advantages and disadvantages of diagnostic methods for the presence of IMNV

Diagnostic method	Advantages	Disadvantages
Histology	• Presumptive diagnosis • Allows an evaluation of infection distribution in various tissues • Less expensive	• Might not detect low-levels of infection • Needs 2–7 days of preparation time • Not specific • Depends on the expertise of pathologists
Transmission electron microscopy (TEM)	• Provides powerful magnifications and detailed information on the structure of IMNV	• TEM is required • Laborious and technically complex • Expensive • Not a sensitive diagnostic tool
RT-PCR (and RT-qPCR) and RT-LAMP	• Most sensitive • Can be used to test all life stages • IMNV specific • Rapid results	• Easy to have contamination problems • Technically complex • RT-LAMP results can be highly variable at low levels of infection
<i>In situ</i> hybridization (ISH)	• Specific	• Histological preparation of tissues required • Laborious • Need to prepare an IMNV-probe
Antibody-based assay	• Specific • Can be developed for pond-site diagnostics	• Generating IMNV-antibodies is expensive and time consuming • IHC is a laborious method • Not a sensitive method
Rapid methods from commercial kits	• Rapid diagnostic results • Field ready • Sensitive RT-qPCR-based tests	• Need to purchase from commercial sources • May be expensive • Can only run limited numbers of samples, especially with pond-site devices
Bioassay	• Demonstrates the presence of viable IMN virions	• Requires fresh (or frozen) infected tissues • Takes a period of 7–30 days • Requires follow-up confirmations with other methods • Need a wet-laboratory facility and healthy indicator shrimp • Expensive to run

In a suspected IMN outbreak, RT-PCR (RT-qPCR) should be used as the initial confirmatory test as it provides a rapid turnaround. The validity of the RT-PCR (RT-qPCR)-positive data should then be confirmed by histological evaluations and associated methods such as ISH or IHC. A definitive association can be made from the presence of IMNV determined by RT-PCR (RT-qPCR), histopathological change, and laboratory infection studies.

4. PREVENTION AND TREATMENT

4.1 Vaccination

There are no effective vaccines for IMNV because the shrimp immune system has no memory and, therefore, is not capable of acquired immunity. However, it is suggested that RNA interference (RNAi) triggered by sequence-specific dsRNA can function as a “vaccine” to prevent IMNV infection and disease (Bartholomay *et al.*, 2012). RNAi is a specific cellular response based on recognition and specific cleavage of exogenous dsRNA. It will destroy viral messenger RNA (mRNA) specifically, and thus inhibit virus replication (Robalino *et al.*, 2007; Hirono *et al.*, 2011). RNAi has rapidly become a powerful tool for treatment of shrimp diseases. Several studies on the use of dsRNA to trigger the RNAi pathway in shrimp have reported increased survival against shrimp viruses, e.g. white spot syndrome virus (WSSV) (Xu, Han and Zhang, 2007; Mejía-Ruiz *et al.*, 2011), yellow head virus (YHV) (Yodmuang *et al.*, 2006) and Taura syndrome virus (TSV) (Ongvarrasopone *et al.*, 2011).

In laboratory trials, *P. vannamei* that were treated with IMNV-specific dsRNA survived an initial infection and were resistant to second viral challenge 50 days later (Loy *et al.*, 2012). Further studies also showed that administration of dsRNA (0.5–5 µg per shrimp) two days prior to IMNV challenge can significantly lower the mortalities from IMN (Feijo *et al.*, 2015). Moreover, these dsRNA constructs can be utilized as a therapeutic treatment against IMNV when administered post-infection. In laboratory trials, administration of 0.5 µg/shrimp of IMNV-dsRNA5 at 48 hr following IMNV exposure resulted in over 50 percent survival compared to other treatment groups (Loy *et al.*, 2013).

This dsRNA-based protection has shown tremendous potential from the results of the laboratory studies. Research on delivery methods is ongoing for such approaches to be applied in the field. One example is using biodegradable nanoparticles as a delivery platform for encapsulation and release of IMNV-dsRNA. In the laboratory two-week challenge studies, the use of nanoparticles encapsulation to deliver IMNV-dsRNA (100 µg) to *P. vannamei* demonstrated a 70 percent survival, an equivalent survival as injecting dsRNA (Phanse *et al.*, 2015). The hydrophobic polyanhydrides of the nanoparticles can prevent the exposure of encapsulation payloads to water, therefore enhancing their stability. In addition, this nanovaccine platform is shown to be capable of “immunization” of shrimp via immersion or feeding (incorporation into shrimp feed) and can be used under farm conditions (Chalamcherla, 2015).

This RNAi technology is shown to improve shrimp survival in IMNV laboratory challenge studies; but so far, there are no treatment methods available to prevent or control outbreaks of IMNV in the field.

4.2 Immunity and resistance

Shrimp have an innate immune system against viral infection, which involves only non-specific, physiological responses. These include humeral responses involving Toll-like receptors (Yang *et al.*, 2007), a prophenoloxidase (proPO) activating system, a clotting cascade (Lai, Chen and Kuo,

2005; Ai *et al.*, 2008, 2009) and production of antimicrobial peptides (e.g. antioxidant enzymes and lectins). Cellular immune responses involving apoptosis, encapsulation, phagocytosis and melanization are also known (Iwanaga and Lee, 2005; Little, Hultmark and Andrew, 2005; Tassanakajon *et al.*, 2013). The JAK-STAT signaling pathway, which responds to extracellular chemical signals, has also been shown to play a role in a response to viral infections of shrimp (Robalino *et al.*, 2004, 2005; Chen *et al.*, 2008).

Enhancement of disease resistance has been attempted through stimulation of the innate immunological system, such as prophenoloxidase responses, by inclusion of β -1,3/1,6-glucan (0.1 percent) in the shrimp feed (Neto and Nunes, 2015).

5. EPIDEMIOLOGY

5.1 Geographic distribution

IMNV has so far been restricted to farmed shrimp in Brazil, Indonesia, India and Malaysia.

In **Brazil**, the IMN outbreaks occurred during August 2002 to March 2003. Initially, IMN impacted ponds of a large farm located at Mexeriqueira (Piauí), then other farms located in the states of Piauí, Ceará and Rio Grande do Norte. Mortalities were observed due to IMN from nurseries and grow-out phases of culture (Thales Andrade, unpublished data). In the IMNV-affected ponds, the mortalities ranged from 55 to 79 percent, and the feed conversion ratio (FCR) increased from 1.5 to 4.4. At present (2018), the disease has been spread to all the states that practice shrimp farming, extending from the State of Paraíba to Santa Catarina, passing through the states of Pernambuco, Alagoas, Sergipe and Bahia. In the recent years, the IMN outbreaks have been less problematic, with lower mortalities; it is not clear if the shrimp has developed resistance to this disease (M.L. Santos, personal communication, March 14, 2018).

In **Indonesia**, within a few years, IMNV had spread from South America across the Pacific to the island nation of Indonesia. It first appeared there in 2006, being reported from several farms clustered in Situbondo District on the Island of Java; disease occurred in *P. vannamei* stocked after 30–90 days and mortalities ranged from 10 to 30 percent (Taukhid and Nuraini, 2009). From there it quickly spread throughout the archipelago reaching Bali and the Lesser Sunda Islands the following year and the island of Sumatra by 2008. The disease was reported in Central Java, West Java and North Sumatra provinces and across the Java Sea in West Kalimantan and South Sulawesi provinces by 2009 (Naim, Brown and Nibert, 2014). Eradication of IMNV from these regions has proven problematic, so producers have tried to cope with the continued presence of the disease, with some success, through the adoption of better management practices (BMPs). In 2017–2018, the progression of the disease became rapid (shrimp died off within in a week), with a higher mortality (20–40 percent) as compared to previous years. These moribund/dead shrimp were severely infected with IMNV. It is not clear if the virus has evolved and become more virulent, or if the shrimp were co-infected with other pathogens such as WSSV and the microsporidian *Enterocytozoon hepatopenaei* (EHP), resulting in an increased mortality rate (H. Bambang, personal communication, March 20, 2018).

In **India**, IMNV was detected in farmed *P. vannamei* in the Purba Medinipur District, West Bengal State in July 2016. The infected shrimp weighed 12.0 to 12.3 g at the age of 55–57 days; the mortalities ranged from 20 to 50 percent. The diseased shrimp exhibited gross signs characteristic of IMN, and IMNV infection was confirmed by RT-PCR and bioassays (Sahul Hameed *et al.*, 2017). Later, in April 2017, IMN was also found in the largest *P. vannamei* farming state, Andhra Pradesh.

In **Malaysia**, IMNV was detected by RT-PCR in a pond within a farm culturing *P. vannamei* in the State of Melaka in June 2018 (OIE, 2018c).

When IMNV first emerged, it resulted in high mortalities for a few years before the disease events became more progressive. In these later cases, the pathology of infected shrimp was often chronic in nature. However, the viruses carried in chronically infected shrimp remain pathogenic to healthy shrimp, and infection in chronically infected shrimp can manifest as an acute infection under various conditions of stress. Stress factors can arise from many sources, including water temperature, salinity, stocking density or pond biomass.

5.2 Persistence in the environment

No data are available regarding persistence of IMNV in water and pond sediments or on farm equipment.

5.3 Modes of transmission

IMNV can be transmitted both horizontally and vertically. From *per os* bioassays, it is clear that individuals can become infected through ingesting infected tissues. It is likely, although there is no definitive evidence, that the virus can be transmitted via the water or sediment, as has been demonstrated with other shrimp viruses. IMNV can be transmitted via ingestion of infected tissue. Once IMNV is established in a pond, the virus can be transmitted rapidly through cannibalism of sick and dead shrimp. Transmission via cannibalism is supported by feeding trials in which groups of shrimp ingesting infected tissue soon developed infection with IMNV and suffered mortalities. Shrimp larvae can become infected through vertical transmission. IMNV has been detected in the eggs and ovaries of infected females, suggesting that maternal transmission is involved (da Silva *et al.*, 2016).

5.4 Vectors and reservoir hosts

The brine shrimp *Artemia* sp. is an essential feed used in shrimp larviculture and maturation. Through laboratory infections, it has been shown that *A. franciscana* can act as a mechanical vector for IMNV (da Silva *et al.*, 2015).

IMNV is a non-enveloped virion and likely can remain infectious in the gut and faeces of seabirds that feed on dead or dying shrimp at farms with on-going IMN epizootics. Then IMN could be

spread within and among farms through bird faeces or regurgitated shrimp carcasses, similar to one of the transmission routes for TSV (Vanpatten, Nunan and Lightner, 2004).

Native wild penaeid shrimp, including *P. subtilis*, in northeastern Brazil have been anecdotally reported as hosts (OIE, 2018b).

5.5 Factors influencing disease transmission and expression

The incidence of IMN usually becomes more frequent during the rainy season in northeastern Brazil. In laboratory infection studies, it has been shown that lowering salinity from 35 ppt (normal condition) to 5 ppt (stress condition) reduces the generation time of IMNV from 57.4 to 25.2 min, thus increasing the proliferation of the virus (Vieira-Girão *et al.*, 2015). This change may be related to osmotic effects on the shrimp immune defense system.

5.6 Impact of the disease

Mortality from IMNV infection can reach 70 percent at harvest time, with over 40 percent mortality occurring late in the production cycle and reduction of market value of survivors that have noticeably necrotic tail muscles. Economic losses during the 2002–2003 IMNV outbreaks in Brazil were estimated at USD 20 million. Brazilian shrimp farmers suffered economic losses of USD 440 million as a result of IMNV outbreaks at the end of 2005. The accumulated losses from both Brazil and Indonesia due to IMNV from 2002 through 2011 were estimated to be more than USD 1 billion (Lightner *et al.*, 2012). Additional costs, not yet quantified, would include costs of facility disinfection, implementing control measures, and the monitoring of stocks.

6. PRINCIPLES OF CONTROL AND ERADICATION

This section provides information relevant to the development of management strategies needed for responding to either severe IMN outbreaks or the detection of subclinical IMNV infections in populations of farmed shrimp, particularly *Penaeus vannamei*. The information is most relevant to the shrimp-farming industries of Brazil, Indonesia, India and Malaysia, but the measures described can also be applied to other countries. No effective treatments are available for IMNV-infected shrimp cultured in farms. The disease, once becoming established, could persist in wild populations of indigenous shrimp, such as *P. subtilis* (in Brazil), *P. monodon* (in Indonesia and India) and *P. merguensis* (in Indonesia). Control measures for these wild populations would not be feasible.

6.1 Methods for control and elimination

There are a number of actions that can be taken to prevent the spread of IMNV, including quarantine and movement controls, tracing investigation, surveillance, the use of IMNV-free PL, the use of disease-resistant shrimp, the use of probiotics, disinfection of shrimp products and eggs/nauplii, emergency harvest, destruction of infected shrimp, disposal of diseased hosts, decontamination of infected farms and vector control. These methods, as well as some environmental considerations and the need for public awareness, are discussed below.

6.1.1 Quarantine and movement controls

Quarantine and movement restrictions should be implemented immediately upon suspicion of an IMN outbreak. The Competent Authority should establish appropriate zone and compartment designations which are management strategies to control, protect shrimp health status and facilitate international trade in shrimp and shrimp products. Zoning relies more on geographic barriers, and is usually under the responsibility of the Competent Authority, while compartments are within the production facilities and managed with biosecurity programmes to maintain the health status. Detailed information regarding zoning and compartmentalization can be found in chapter 4.1 of the Aquatic code (OIE, 2018a).

For zoning, zones can be designated as (see Figure 7):

- Infected premises or area: the premises (e.g. farm) or area (e.g. geographically separated area) where the infection is present, and its immediate vicinity. This premise is where a presumptive positive case or a confirmed positive case exists based on laboratory results.
- Buffer zone: an area adjacent to infected premises or area, which is susceptible to become an infected area.
- Free area: a non-infected area.

A control area consists of infected premises and buffer zone.

Movement controls from the infected premises (area) should include:

- Bans on the movement of live and uncooked (raw) shrimp from the infected premises into IMNV-free areas and off-site processing plants
- Restrictions or bans on releasing shrimp and pond water from the infected premises into aquatic environments
- Bans on using uncooked shrimp in the infected premises as baits for fishing
- Bans on discharging of processing-plant effluent without any treatment within the infected premises
- Control of disposal of dead shrimp
- Control of seabird access to live and dead shrimp within the infected premises
- Restrictions or bans on the use and movement of equipment and vehicles between farms within the infected premises

The implementing of these bans will depend on the severity of the disease, the types of operation (such as farm location, farm size, pond system), and the response options chosen. For a fish virus, infectious salmon anemia virus, a control area of 5 to 6 km is recommended (Mardones *et al.*, 2011).

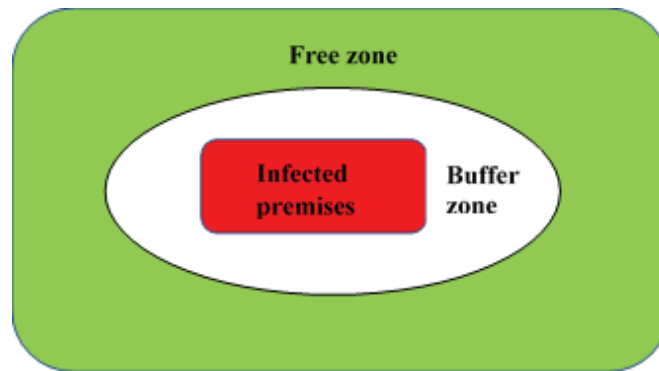


Figure 7. Relationship of free zone, buffer zone and infected premises in the infectious myonecrosis (IMN) outbreak response.

6.1.2 Tracing

Tracing refers to investigation of: (a) if the infected shrimp were moved to other areas, (b) if the IMNV infection has spread to other areas, and (c) the sources of the disease. Movements of the following from infected sites or premises might need to be traced:

- live shrimp: e.g. broodstock, PL and other stocks, including those sold through bait shops;
- uncooked (fresh and frozen) shrimp intended for human consumption or for bait;
- live feed: *Artemia* biomass;
- effluent and waste products from processing plants and farms: discharge into or nearby coastal or inland waters; and
- vehicles (transport vehicles, feed trucks, cars, boats) and farm materials (nets, buckets and other farm equipment).

IMNV is a rapidly evolving dsRNA virus; the genomic sequencing and phylogenetic analysis have proved that Brazilian and Indonesian IMNV have a common origin. Whole genome sequencing (WGS) has been shown to have a higher resolution for differentiating among isolates. Thus, it is suggested a WGS database for the IMNV isolates be developed. This will increase the capability to trace, confirm the origin of and determine the relatedness of outbreaks.

6.1.3 Surveillance

Surveillance is the periodic sampling of populations and screening for the presence of IMNV. It can be used to detect the early occurrence of IMN and determine the prevalence of IMNV in populations and is used in the process of maintaining and certification of farms or areas for freedom

from IMNV. Detailed information on the general requirements for surveillance to establish freedom of IMNV at various prevalence thresholds is provided in chapter 1.4 of the Aquatic code (OIE, 2018a).

Shrimp samples to be tested for IMNV can be groups of individuals, PL, pleopods collected from juveniles, or larger shrimp. The samples can be frozen, or preserved in ethanol or in other transportation solution, depending on the requirements of the diagnostic laboratory to which they will be sent. Depending on the information needed, samples can be pooled prior to diagnostic analysis to reduce costs. Surveillance could also be used to examine the presence of IMNV in wild populations, but statistical protocols for such studies would need to be developed on a case-by-case basis.

Methods for the presumptive and confirmatory diagnosis of IMNV are presented in Section 3.

6.1.4 Use of IMNV-free postlarvae

In most shrimp-farming countries, including Indonesia, the hatcheries rearing *P. vannamei* are mostly using IMNV-free broodstock and implementing strict biosecurity to produce PL free from IMNV. IMNV can be transmitted to PL by vertical and horizontal routes. To make sure that the broodstock and PL are free from IMNV, the manager regularly checks for IMNV and other serious diseases using RT-PCR (or RT-qPCR) methods. If a positive result is found, then IMNV infection should be confirmed by a national reference laboratory. If the results of the national reference laboratory confirm positive infection, the PL are not allowed to be stocked into grow-out ponds. The IMNV-positive PL should be destroyed and the entire hatchery disinfected by use of chemicals such as chlorine, followed by drying for two weeks to ensure eradication of the virus.

6.1.5 Use of IMN-resistant shrimp

Early strategies to reduce the risk of IMN to farms focused on the use of specific-pathogen-free (SPF) stocks; however, as IMNV became established in the environments surrounding shrimp farms, interest shifted to the use of specific-pathogen (i.e. IMNV)-resistant (SPR, resistant to IMN disease, tolerant to IMNV infection) stocks.

In a laboratory challenge for identification of IMN-resistant family lines of *P. vannamei* developed in Hawaii, United States of America and subsequent breeding of resistant broodstock, 21 family lines were challenged for a minimum of 20 days, and the results showed significant improvement in survival (reaching to 95 percent) (White-Noble *et al.*, 2010). Another selective breeding for IMN resistance was initiated in Brazil; under controlled conditions of challenge with IMNV for 60 days, mean survival rate increased from 3.2 percent in the F1 generation to 55.3 percent in the F7 (Lima *et al.*, 2013).

In laboratory challenge studies with other penaeid species, no mortalities were observed in either *P. stylirostris* or *P. monodon* (Tang *et al.*, 2005). Hence, restocking of ponds with *P. monodon* or *P. stylirostris* can possibly reduce losses due to IMN.

Selection for disease-resistant shrimp populations has been successful for TSV in *P. vannamei* (Moss, Doyle and Lightner, 2005), for infectious hypodermal and haematopoietic necrosis virus (IHHNV) in *P. stylirostris* (Tang *et al.*, 2000) and for WSSV in *P. vannamei* (Cuéllar-Anjel *et al.*, 2012). The disease-resistant shrimp become tolerant to viral infection, and do not suffer mortalities from disease outbreaks.

The development of TSV-resistant shrimp is an example of the successful use of a disease-resistant shrimp line under farm conditions, and has been a great benefit to improve production and profitability for those farmers who experience outbreaks of Taura syndrome (TS disease). In recent years, the mean survival of the best-performing families has reached 100 percent in laboratory challenge studies. Since the introduction of the TSV-resistant shrimp line in the shrimp farming industry, TS seems to have disappeared.

6.1.6 Use of probiotics

Probiotics are shown to enhance the innate immunity (non-specific immunity) in shrimp and defend against pathogens, including viruses and bacteria. The addition of probiotics as antiviral agents is very common in the shrimp farming industry. For example, *Bacillus megaterium* is shown to increase the resistance to WSSV and vibriosis in *P. vannamei* (Balcazar, 2003; Li, Tan and Mai, 2009).

Another role for probiotics is to improve the water quality (i.e. pH, dissolved oxygen, dissolved carbon dioxide and organic loads and minerals). Bacteria, such as *Bacillus* sp., have been shown to convert organic matter into carbon dioxide efficiently. In addition, probiotics are beneficial in modifying water quality, as they can increase the composition of microbial species (Mohapatra *et al.*, 2013). The pH, dissolved oxygen, NH₃ and H₂S in rearing water were found to be of a higher quality when probiotics were added, thus maintaining a healthy environment for shrimp (Banerjee *et al.*, 2010).

Shrimp fed a synbiotic containing the probiotic bacterium *Vibrio alginolyticus* and a prebiotic (sweet potato oligosaccharides) were shown to have a higher survival and better growth than the control groups in laboratory IMNV-challenge trials (Oktaviana, Widanarni and Yuhana, 2014). Shrimp fed this synbiotic had an improved immunity as demonstrated in phenoloxidase activities, haemocyte counts and respiratory burst parameters.

In Indonesia, farmers usually apply probiotics twice per week to improve the water quality. The probiotics used are nitrifying bacteria, *Bacillus* spp. and photosynthetic bacteria. When disease is detected in the shrimp population, *Bacillus* spp. and *Lactobacillus* spp. are mixed with shrimp feed to enhance resistance to disease. Indonesian government regulation will not allow the use of probiotics containing more than five species of bacteria as a registered probiotic.

6.1.7 Disinfection of shrimp products and eggs/nauplii

IMNV is a stable virus and can remain infectious for extended periods in frozen shrimp prior to their sale in retail outlets. When IMNV is present in tissues, host protein mass will protect viral

infectivity against thermal destruction. Because of this, biological wastes derived from IMNV-infected shrimp should be heated at 60 °C for more than 20 min (based on the studies from WSSV, e.g. Balasubramanian *et al.*, 2006) to ensure that the virus is inactivated.

In shrimp hatcheries, IMNV can be transmitted to and among offspring by vertical and horizontal routes. Washing eggs/nauplii extensively in sea water can be beneficial to reduce viral loads and the prevalence of IMNV-infected shrimp. Methods of eggs/nauplii disinfection are detailed in chapter 4.4 of the Aquatic code (OIE, 2018a).

6.1.8 Emergency harvest

Within the infected premises, if shrimp are grown to a sufficient size, they can be emergency harvested, cooked and sold for human consumption. Emergency harvest is a common practice for farmers to recover some economic losses from the outbreaks. However, food-safety standards and virus inactivation requirements must be met.

6.1.9 Destruction of infected shrimp

Any chemical agents used to disinfect or destroy IMNV-infected animals must be approved for that use by the Competent Authority. The relevant state or local Competent Authority should also be consulted for advice before the use of such chemicals.

Slaughter of diseased shrimp should be both hygienic and humane, and avoid spillage or escape to the environment. Scavengers such as seabirds should be kept away through the use of electric fence barriers or netting to prevent the mechanical spread of IMNV during the destruction process (see section 6.1.12).

6.1.10 Disposal of diseased hosts

IMNV-infected moribund and dead shrimp are infectious. Therefore, they and other possible infectious wastes are sources of infection, such as potential aquatic carriers, must be destroyed and disposed of immediately and appropriately to reduce risks of spread. Burial is least expensive and is a practical method for disposing of large amounts of diseased shrimp. To reduce the risk of infectious material entering natural waters or the exposure of susceptible species, a burial site should be selected that is remote from shrimp culture areas. Heat effectively destroys all viral pathogens; thus, incineration can be used. An incinerator must be capable of handling the water content involved. See chapter 4.7 of the Aquatic code (OIE, 2018a) for details of disposal methods.

6.1.11 Decontamination of infected farms

Decontamination of a farm affected by IMNV involves destruction of stocks and disinfection of all components of the facility. Viruses can remain viable for several days in pond water. Thus, the water should be disinfected through the addition of chlorine to a concentration of 50 parts per

million (ppm), with the water held for four days prior to discharge. Dead shrimp should be removed prior to disinfecting the pond water. Pond sediments should then be sun dried thoroughly for at least two months and may also be treated with a minimum of 0.5 kg/m² of lime (CaOH₂). Equipment (i.e. nets, boots, aerators, pipes, etc.), tanks and building surfaces need to be disinfected (see Chapter 4.3 of the Aquatic code (OIE, 2018a) for details of disinfection methods).

6.1.12 Vector control

Seabirds are known to be important vectors for shrimp viruses. They are commonly found around shrimp farms, where they feed on the dead or moribund shrimp at the pond edges. They often disgorge infected shrimp in other ponds, thus spreading the viruses to uninfected ponds. In addition, the scavenging seabirds defecate in the surrounding waters and ponds, thus contaminating unaffected areas. Therefore, access of seabirds to IMNV-infected shrimp ponds must be controlled. Netting the farm sites and the use of deterrents such as polythene bags, thermocol and burning crackers are effective in this regard. Where possible, contact between crabs and IMNV-infected farmed shrimp should also be prevented. Crab intrusion can be prevented by building fences around the farm sites.

6.1.13 Environmental considerations

A management plan that addresses transmission of IMNV to wild populations should be developed as soon as possible after identification of IMNV infection in the farmed shrimp. An assessment will require information on the density and distribution of the susceptible species and reservoirs. Appropriate management principles must be applied to reduce the chance of diseased shrimp escaping from farms to the wild. Any environmental impacts associated with the destruction and disposal of infected shrimp or materials should be minimized.

6.1.14 Public awareness

In order to reduce the potential of IMNV spreading to other areas, information regarding the disease and its management must be made available to the general public. IMNV is included on the list of shrimp viruses that are required to be reported to the OIE. Industry professionals will need to know that live or frozen shrimp imports, whether from local, regional or international sources, from infected areas need to be avoided. Producers in neighbouring areas need to know what steps to take to prevent the spread of IMNV to their facilities.

Information regarding the IMNV outbreak status and actions being taken to control and eradicate the disease can be disseminated in a variety of forms, such as through extension services, public meetings with shrimp growers associations and the general public, distribution of information and communication technology (ICT) materials such as pamphlets and information sheets by extension officers, and through newspaper articles and radio broadcasts, as well as through more technical routes, such as academic publications, agency technical reports, industry bulletins, and aquaculture workshops and symposia. Social media and other appropriate mobile applications can also be used.

6.2 Control, containment and eradication options

The feasibility of containing an IMN outbreak will depend on prompt diagnosis, the extent of the outbreak and the rapidity of response. Eradication of IMNV from a country will be possible if the virus has not spread to wild populations of shrimp or other aquatic organisms. Eradication is most likely to be successful if the outbreak is localized. In areas where IMNV has become established in wild populations, country-wide eradication may not be possible. However, measures can be taken to prevent further spread of the disease through its containment to infected premises (compartments) or geographical areas (zones) where control efforts are continued. Control and mitigation efforts involve the implementation of management practices that reduce the incidence and severity of IMN outbreaks.

6.2.1 Eradication

The complete eradication of IMNV from Brazil is viewed as impractical because of its presence in wild shrimp (e.g. *P. subtilis*, susceptible to IMNV) and the heavy reliance of hatcheries on wild-caught broodstock of *P. vannamei*. However, eradication of this, and other viral diseases, is feasible for those production facilities that are under strict management. Eradication at facilities will entail implementing procedures that have proven effective with other shrimp diseases (such as a WSSV outbreak in the State of Louisiana, United States of America). If possible, the origin of the IMNV infection should be determined, to prevent re-infection of decontaminated areas.

After decontamination and an appropriate period of fallowing of the production facilities (see chapter 4.6 of the Aquatic code, OIE, 2018a), farmers should determine whether or not the process was effective. This can be accomplished by stocking small areas with healthy shrimp that are highly susceptible to IMNV infection. The use of bivalves (e.g. oysters, clams) as sentinels has also been suggested. Although bivalves do not become infected with IMNV, as a result of their filter feeding these organisms may act as mechanical vectors. The presence of the virus can be detected by RT-qPCR. Such sentinel organisms must be acquired from IMNV-free areas. However, protocols for the use of sentinel organisms in aquaculture facilities have not been developed.

Disinfected farms can resume production, provided that strict measures are taken to prevent re-infection, especially through the use of IMNV-free PL or broodstock obtained from reliable sources. After restocking of the production facility, the stocks should be inspected weekly for the appearance of any clinical signs of IMN for a period of at least one month. Then, samples should be taken in accordance with approved protocols (chapter 2.2.5 of the Aquatic manual, OIE, 2018b) and tested for the presence of IMNV by RT-qPCR.

Eradication will require:

- prohibiting imports of potentially IMNV-infected broodstock, PL or other shrimp, either farmed or wild-caught;
- screening imports of live or frozen commodity shrimp, as well as other susceptible species (e.g. *Artemia* sp.) or potential carriers (e.g. bivalves);

- destruction and safe disposal of all shrimp at infected farms; shrimp could be harvested for markets as long as they are processed by cooking (which renders the virus inactive) either on-site or at processing plants; fresh or frozen shrimp should not be marketed as viruses can remain infectious in these products;
- disinfection of pond and reservoir water before discharge, or prevention of discharged water from entering coastal waters through the use of evaporation ponds; and
- decontamination of pond bottoms, tanks, equipment, supplies, etc., and facility surfaces.

6.2.2 Containment and zoning

If viral eradication is not feasible following an outbreak, zoning and associated disease control measures should be implemented to prevent the spread of the virus to IMNV-free areas. The containment of the virus to specific areas can be accomplished through restricting the movement of shrimp products, water or other contaminated items from infected premises and through the establishment of buffer zones. Conceptually, these buffer zones undergo routine surveillance with the intent of detecting any spread of the virus from the infected premises. However, there are no established guidelines for defining or monitoring such buffer zones, so these procedures would have to be developed by the appropriate governmental agency on a case-by-case basis.

The measures that need to be implemented include:

- establishing infected premises and free areas;
- prohibiting movement of infected shrimp and any contaminated materials into IMNV-free areas; and
- establishing well-monitored buffer zones where the spread of IMNV can be detected before the IMNV-free areas are affected.

6.2.3 Control and mitigation of disease

If the disease cannot be eradicated or effectively contained to specific areas, then measures to control IMNV and mitigate production and economic losses will need to be developed. Such measures could include implementing more rigorous methods of eliminating potential vectors such as seabirds, small crustaceans, bivalves and land crabs. Depending on the situation, this may entail covering the production areas with nets, filtering intake water and constructing barriers to prevent the movement of crabs into the ponds. Control could be placed on movements of live shrimp from the infected premises; more stringent systems might need to be implemented to filter intake water.

In addition, outbreaks of IMN may be triggered when the farmed populations are under increased stress; thus, reducing sources of stress may help prevent low-level infections rising to acute levels. Stress could come from a variety of sources such as high stocking densities, poor water quality or other less optimal culture environmental conditions (e.g. temperature and salinity). In both Brazil and Indonesia, IMN outbreaks occur more often during the hot seasons, when the temperature is above 30 °C. Associated issues include fluctuations of the plankton during the day and deterioration of the pond sediments, which produce harmful gases (e.g. H₂S, NH₃) and toxic

materials. To ameliorate these problems, farmers attempt to maintain good pond conditions through increasing aeration, carefully monitoring and adjusting the feeding regime, applying probiotics and siphoning ponds to remove sludge.

6.3 Trade and industry considerations

Areas that have problems with IMNV may suffer economically because international exports of fresh or frozen shrimp products may be curtailed and these commodities limited to local markets. Associations of local and regional shrimp producers have highly invested interests in eliminating any shrimp disease that negatively affects production or product values. Such associations could prove valuable in the design and implementation of biosecurity programmes to prevent IMNV outbreaks and in the development of appropriate response strategies.

6.3.1 Domestic markets

Production from IMNV-affected farms would not be desirable in domestic markets outside of the infected premises unless the products are dried, canned, processed, cooked or value-added (such as breaded, battered, marinated). Agencies dealing in commerce may place restrictions on transporting or marketing fresh or frozen shrimp products between infected and disease-free areas.

6.3.2 Export markets

IMNV is listed by the OIE as a notifiable disease (OIE, 2018a). In countries where IMNV is exotic, import conditions, such as requiring imports to be certified free of IMNV and testing by commerce inspection organizations to reject shrimp batches that test positive are in place. In Brazil, Indonesia, India and Malaysia, the Competent Authority of each country is responsible for issuing health certificates for all exports to the standards specified by the individual importing countries.

The safety of shrimp products is addressed in Chapter 5.4 of the Aquatic Code (OIE, 2018a). The criteria are that: either there is no IMNV present in the products or, if present, that it has been inactivated through processing. To determine if IMNV is present, the products should be analyzed with diagnostic procedures in accordance with Chapter 2.2.5 of the Aquatic Manual (OIE, 2018b). Processing treatments that inactivate the virus include: (i) physical (e.g. high temperature, drying, smoking), (ii) chemical (e.g. iodine, pH, salt, smoke) and (iii) biological (e.g. fermentation) treatments.

Processed shrimp products, such as cooked, dried, canned or smoked shrimp, as well as shelf-stable products, aquaculture feed, etc. would be considered safe. If the processing treatments are in doubt with regard to viral inactivation, bioassays can be performed through feeding the shrimp products in question to healthy (SPF) indicator shrimp in the laboratory for a duration of one month. At the end of the bioassay, if the indicator shrimp are not infected with IMNV, then the shrimp products do not contain infectious viruses.

7. POLICY DEVELOPMENT AND IMPLEMENTATION

7.1 Overall policy

Addressing biosecurity requires significant resources, strong political will and concerted international action and cooperation. National strategic planning for aquatic animal health (AAH) and biosecurity is vital; without it, a country can only react in a piecemeal fashion to new developments in international trade and serious transboundary aquatic animal diseases, and its aquaculture and fisheries sectors will remain vulnerable to new and emerging diseases. FAO encourages Member Countries to develop and formalize national strategies for aquatic animal health (NSAAH) and health management procedures (FAO, 2007) and to use the Progressive Management Pathway (PMP), a step-wise risk management programme used to develop and monitor national strategies for important livestock diseases such as foot-and-mouth disease, African animal trypanosomiasis, peste des petits ruminants and rabies (Bondad-Reantaso *et al.*, 2018; FAO, 2018, 2019). The actions must be risk-based, proactive, collaborative and should adhere to international standards and regional agreements (both obligatory and voluntary), particularly for those countries sharing transboundary waterways. Responsibilities must be shared among key national, regional and international stakeholders from government, the production sector and academia, as well as other players in the value chain, building on each other's strengths towards a common goal. The NSAAH is a basic element of Stage 1 of the PMP for improving aquaculture biosecurity.

A NSAAH is a broad yet comprehensive strategy to build and enhance capacity for the management of national aquatic biosecurity and aquatic animal health. It contains the national action plans at the short-, medium- and long-terms using phased implementation based on national needs and priorities; outlines the programmes and projects that will assist in developing a national approach to overall management of aquatic animal health; and includes an Implementation Plan that identifies the activities that must be accomplished by government, academia and the private sector.

The development of a NSAAH includes a gap analysis (achieved through a self-assessment survey and strengths weaknesses opportunities and threats (SWOT) analysis) conducted by national and regional focal points, a committee or a task force, a working group on aquatic animal health or any structure that fits the country. The working group will have specific terms of reference. The technical elements that may be considered in the strategic framework will vary depending on an individual country's situation, and thus may not include all the programme elements listed below (alternatively, additional programmes may be identified as having national and/or regional importance and thus need to be included):

- i. Policy, Legislation and Enforcement,
- ii. Risk Analysis,
- iii. National Pathogen List,
- iv. Health Certification, Border Inspection and Quarantine,
- v. Disease Diagnostics,
- vi. Farm-level Biosecurity and Health Management,
- vii. Use of Veterinary Drugs and Avoidance of Antimicrobial Resistance (AMR),

- viii. Surveillance, Monitoring and Reporting,
- ix. Communication and Information Systems,
- x. Zoning and Compartmentalization,
- xi. Emergency Preparedness and Contingency Planning,
- xii. Research and Development,
- xiii. Institutional Structure (including Infrastructure),
- xiv. Human Resources and Institutional Capacity, and
- xv. Regional and International Cooperation.

It provides clear objectives for all relevant activities, defines the activities that need to be accomplished to reach these objectives, and gives an indicative time frame and priority for each activity. Its development involves an extensive process during which the current national aquatic animal health capacity and future goals are first assessed; and then, policies, priorities and needs are identified. An iterative process, it involves the national Competent Authority and other relevant authorities and extensive consultation with key stakeholders from other government agencies, academia and the private sector. Special attention to the needs and empowerment of small-scale producers should be accorded priority, as they represent the weak link in any biosecurity system. IMN can thus be included in a country's National Pathogen List if it satisfies the country's criteria for disease listing and considered in all biosecurity measures following the above-mentioned elements of a NSAAH (e.g. IMN diagnostics, IMN surveillance and reporting to OIE, IMN zoning, IMN disease strategy manual, etc.).

7.2 IMN-specific objectives

Following the initial diagnosis, the head of the Competent Authority and/or the Chief Veterinary Officer should select the most appropriate control options for responding to outbreaks of IMNV: option 1 – eradication of IMNV from the shrimp farms; option 2 – containment and zoning; or option 3 – control and mitigation of IMN. The aims are:

- to eliminate IMNV from the country when possible;
- to prevent re-emergence of the IMN disease;
- to prevent the spread of the disease to farmed or wild populations outside of infected premises;
- to minimize the economic impact of the disease on commercial production;
- to prevent losses of domestic and international markets for locally farmed shrimp; and
- to ensure that stakeholders and the public are informed of the issues involved in preventing the inadvertent introduction or spread of IMNV through improper importation or movement of shrimp products.

7.3 Problems that need to be addressed

An outbreak of IMN in commercial shrimp farms can result in severe economic losses due to high mortality in infected populations and subsequent reduced production. For example, losses from IMN outbreaks on farms in Brazil between 2002 and 2005 were estimated at USD 440 million. If production is depressed, farmers may fall into debt and be unable to pay for the lease of ponds. Also, if shrimp farms are abandoned because of diseases, the mangroves that were altered to construct them may take over 20 years to recover. In addition, shrimp-farm activities that have

resulted in the salinization of adjacent soils may prevent the return of these lands to agricultural use upon the abandonment of shrimp farming. Thus, disease outbreaks can be especially devastating to large areas that support communities depending almost entirely on incomes derived from shrimp farming. Such communities will be placed in jeopardy if IMN outbreaks are not prevented or mitigated through proper management. Responses to the emergence of IMNV must be rapid to mitigate the negative impacts on the shrimp aquaculture industry and prevent the possibility of the virus becoming established in wild populations. Initial efforts should be to identify the IMNV quickly through routine diagnostic procedures and then to institute procedures immediately to eradicate and prevent the spread of the disease.

7.4 Overview of response options

Responses can range from efforts to eradicate the disease from the country to the development of strategies for individual farms to manage production with potential low levels of infection in their stocks. The actions are detailed in the following chart (Figure 8). Eradication strategies can usually only be successful soon after the discovery of introduced IMNV. Containment of the disease to certain zones may be possible in some instances, especially in countries where production facilities are widely separated, such as on different islands in Indonesia. However, if the virus is enzootic or becomes so through spread from farmed to wild populations, then management at the farm level will be the only effective option to reduce economic losses from the disease.

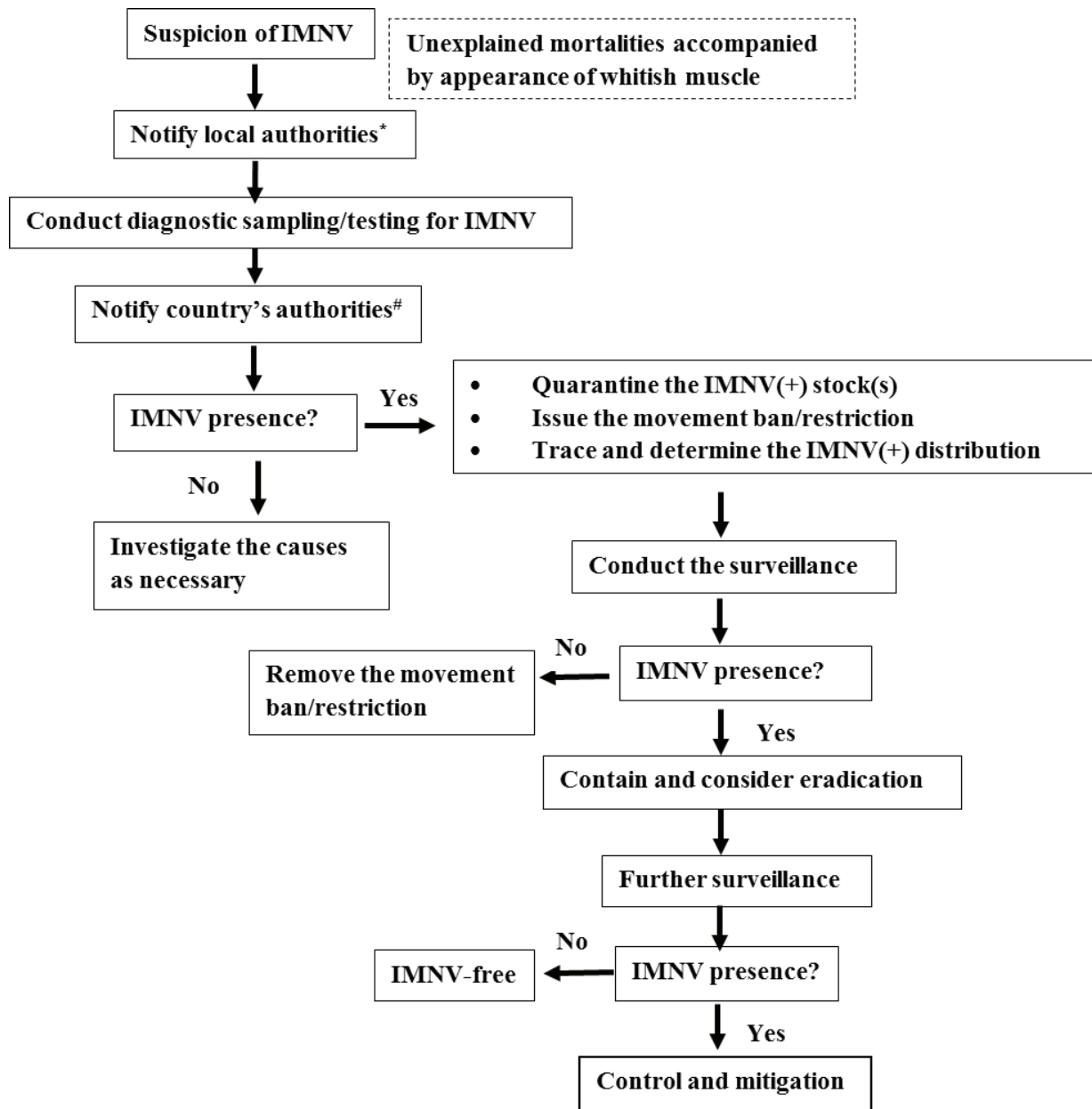


Figure 8. IMNV responses flow chart. *: Aquatic animal health specialists, Aquatic veterinarians or Aquatic animal health professionals; #: Fisheries and aquaculture Competent Authorities and Veterinary Services

7.5 Strategies for eradication and control

7.5.1 Eradication from production facilities

There is now a long history in the global aquaculture industry of responding to shrimp viral diseases. Producers, government agencies and industry organizations have developed and successfully employed protocols for eradication of viruses from shrimp farms. Many producers have met this challenge and have remained active in spite of the on-going costs of maintaining disease-free status.

- **Diagnosis and surveillance** – Obtaining an accurate diagnosis is the first step in disease management. Diagnosis requires samples of tissues from moribund shrimp or live shrimp with clinical signs of disease. Specimens and tissue samples are sent to a competent diagnostic laboratory. Details on collection and processing of samples are given in section 3.4 of this manual and in chapter 2.2.5 of the Aquatic manual (OIE, 2018b).
- **Disposal of diseased stock and disinfection** – To eradicate IMNV from the facility, all diseased stocks must be removed by emergency harvesting or destroyed. Disposal of hosts and decontamination of the infected farm can be found in the sections 6.1.10 and 6.1.11.
- **Restocking with SPF and/or SPR stocks** – Once the facility is disinfected, it can be restocked with stocks certified to be SPF (IMNV-free)/SPR (IMN-resistant). Strict management to prevent re-introduction of IMNV then needs to be implemented.
- **IMNV-free declaration** – There is an OIE declaration procedure for IMNV-free country, zone or compartment. The pathways for these recognitions are listed in chapter 1.4, article 1.4.6 of the Aquatic code (OIE, 2018a). A country, zone or compartment may take a self-declaration of freedom from IMNV if:
 1. **Absence of susceptible species:** There are no susceptible species (e.g. *P. vannamei*, *P. monodon*) present.
 2. **Historically free:** Susceptible species are present, but the area is considered free of IMNV provided that:
 - a. either there has never been a substantiated occurrence of IMNV in any (i.e. farmed or wild) populations, or the disease has not been detected for at least ten years and basic biosecurity conditions have been continuously met during that time; or
 - b. a susceptible species is introduced from an IMNV-free area where basic biosecurity conditions are in place and that previously had no susceptible species present.
 3. **IMNV has been eliminated:** This would apply if susceptible species are present and either an IMNV incident occurred within the previous 10 years or the current disease status is unknown. Proof of IMNV-free status should be based on a series of surveys of farmed and wild populations of susceptible species over at least a two-year period. In addition, basic biosecurity conditions should be in force continuously for the past two years. Surveys should be for at least two years and follow the surveillance protocols recommended in the Aquatic manual (OIE, 2018b). There should be two surveys per year to be conducted three or more months apart. These should be designed to provide a greater than 95 percent confidence with a prevalence of 2 percent or lower. Shrimp to be sampled are preferred to display any clinical signs, such as whitish muscle, red discolouration, lethargy, etc. Such surveys should not be based on a voluntary submission; they should be conducted with the involvement of the Competent Authority of the country.

An example of an IMNV-free compartment can be found in Indonesia, where the shrimp farm PT Bibit Unggul (Global Gen) was declared to be IMNV-free following the OIE guideline described above. This company is located on the Island of Lombok, Nusa Tenggara Barat Province. Starting in 2007, a basic biosecurity protocol was put in place by PT Bibit Unggul and in 2008, a pathogen-surveillance programme was initiated. Health screenings were conducted twice a year for major penaeid shrimp diseases, in accordance with the Aquatic manual (OIE, 2018b). Population sampling was conducted for PCR testing, assuming a pathogen prevalence of 2 percent. None of

the OIE-listed crustacean diseases, including IMNV, were detected through PCR testing. In 2010–2012, the PT Bibit Unggul facilities were inspected by the Delegate of Indonesia to the OIE and certified as meeting both the biosecurity requirement and the requirement that IMNV had not been detected in over two years. Subsequently, PT Bibit Unggul initiated a meeting and discussion between the competent authorities of Indonesia and Viet Nam, resulting in an agreement between the two countries on health screening, quarantine and health certification requirements for the export of *P. vannamei* from Indonesia to Viet Nam; and the Vietnamese Competent Authority has officially approved PT Bibit Unggul as a supplier of *P. vannamei*.

7.5.2 Containment and movement control

It may be possible to contain the IMNV outbreak to certain areas or zones. This may be effective under some circumstances where IMN-free and infected production facilities are not in close proximity. It is a strategy that could be considered as efforts continue to eradicate the disease from infected areas. Examples of this strategy being effectively used to control shrimp diseases over a long period of time are difficult to come by; however, the basic aim of this strategy is to restrict the spread of IMNV. Appropriate government regulations will need to be developed in support of this strategy. The strategies are:

- *Restrictions on movement of shrimp products:* If an IMN outbreak occurs, restrictions on the movements of uncooked shrimp, whether live, fresh or frozen, into IMNV-free areas should be immediately implemented.
- *Restrictions on water discharge:* Water from infected ponds should not be discharged into local coastal or river systems. Use of zero or low-discharge production systems can reduce this hazard.
- *Prevention of spread by seabirds or other wildlife:* This will entail the proper disposal of dead shrimp with care to prevent access to the infected carcasses by seabirds, crabs or other scavengers.
- *Surveillance:* To make sure that the virus is contained, continued surveillance of stocks in areas adjacent to infected premises would be prudent. Detailed protocols for such large-scale, continuing surveillance programmes, however, have not been developed. Statistics regarding sample sizes required to detect shrimp viruses are described in chapter 1.4 of the Aquatic code (OIE, 2018a). The sample sizes depend on the sizes of the populations being monitored and the expected prevalence (i.e. percentage of the population that is infected) of the virus. If IMNV is to be detected before reaching high prevalence, larger samples sizes will be required.

7.5.3 Management and mitigation

If IMNV is enzootic or enters wild populations, efforts of wide-spread eradication or containment are likely to fail. Producers may still operate successfully if farms are managed properly. Two scenarios for successful production under these circumstances are to: i) restock farms where IMN has been eradicated and that have been disinfected; and ii) manage farms that have stocks with chronic IMNV infection. Farms in the first category can, through continued surveillance over a period of two years without re-emergence of the disease, be considered IMNV-free. Farms in the

second category can continue to produce shrimp, but may have to limit exports to areas/countries where IMNV is not a concern.

- *Management of facilities restored to IMNV-free status:* Farms that have been restored to IMNV-free status will have to implement strict management protocols in order to maintain this status. This will entail stocking the facility with SPF/SPR (IMN-resistant) shrimp and ensuring high levels of biosecurity. The basic elements include the physical, chemical and biological methods necessary to protect the shrimp production facility from the diseases that represent a high risk. These elements can be found in the chapter 4.2 of the Aquatic code (OIE, 2018a). In general, a biosecurity programme includes:
 - establishing physical structures and barriers, such as fences and gates, and ensuring that all buildings are secured to prevent unauthorized entrance;
 - having procedures in place to control visitors' activities (e.g. requiring sign in/out, wearing protective clothes, disinfection of footwear, providing hand wash stations, etc.);
 - refusing entry to individuals who have been at a disease-affected facility within the past three days;
 - disinfecting intake water supplies with multiple filters and ultraviolet light (UV) (or ozone) to inactivate shrimp viruses, including IMNV;
 - implementing sanitation procedures in the facility, such as avoiding the movement of equipment (i.e. nets, feed buckets, sampling jars, etc.) from one tank (or pond) to another;
 - cleaning and disinfecting (e.g. chlorine for 30 min) the rearing units (tanks, raceways) before and after production cycles;
 - excluding IMNV vectors during pond preparation;
 - using quarantine and SPF/SPR (IMNV-free, IMN-resistant) certified stocks;
 - not using live or fresh feed from infected premises; and
 - performing routine health inspections of all stocks.
- *Mitigation:* Farms that are not IMNV-free will have to develop management protocols that prevent IMNV from becoming problematic. These management issues should be aimed at reducing other forms of stress to farmed stocks such as maintaining optimal environmental conditions, ensuring high water quality, and the use of IMN-resistant stocks. For *P. vannamei*, optimal temperatures for grow out range from 25 to 35 °C (Ponce-Palafox, Martinez-Palacios and Ross, 1997) and optimal salinities range from 15 to 25 ppt (Boyd, 1989). Water quality can be maintained by water exchanges, careful feeding practices, pond aeration and the application of probiotics. IMN-resistant shrimp lines have been developed and used successfully in Brazil. Production in Brazil was severely impacted by IMNV outbreaks in 2002 thru 2005, however, through development of best management practices, production has returned to levels that were common prior to the outbreaks.

7.6 Capacity building

In order to improve the accuracy of detecting IMNV, the Government of Indonesia has facilitated diagnostic laboratories by providing IMNV RT-PCR and RT-qPCR reagents and associated equipment for screening and confirmatory analyses. Training on identification, detection and diagnosis of IMNV following the methods outlined in the OIE manual was conducted to improve the capabilities and knowledge of the laboratory personnel. Screening and confirmatory laboratories have been designated for hatcheries and grow-out farms.

To control IMNV, Indonesia has collaborated with FAO to conduct the project TCP/INS/3402: Development of preventive aquatic animal health protection plan and enhancing emergency response capacities to shrimp disease outbreaks in Indonesia. The outputs of the project are:

- (1) National Strategy for Aquatic Animal Health (NSAAH);
- (2) Guideline for Emergency Responses and Contingency Plan;
- (3) Guideline for Surveillance on Shrimp Diseases;
- (4) Guideline for Biosecurity on Farm Level; and
- (5) Guideline for Information System.

During the implementation of these policies, national workshops have been conducted and attended by aquatic animal health professionals and stakeholders from industry and academia to improve their knowledge and capabilities to control IMNV in Indonesia.

7.7 Funding and compensation

Contingency planning requires prior commitment of adequate funding by government so that immediate actions can be taken as soon as a disease emergency is recognized. Expenses for various aspects of the response to an IMN outbreak can be substantial, the individual producer and shrimp-grower associations would fund some of which, while others may be more appropriate for funding through regional or national governments.

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Annex 1

Country delegates and resource experts to the FAO Project TCP/INT/3501:
Strengthening biosecurity governance and capacities for dealing with the serious shrimp
infectious myonecrosis virus (IMNV) disease

Country	Delegates
Brazil	Marcelo LIMA SANTOS Rodrigo ROUBACH
People's Republic of China	Jie HUANG Yan LIANG
Ecuador	Ulbio Alcides PAREDES NIETO Narciso PIN QUÍMIZ
Indonesia	Hanggono BAMBANG Mukti SRI HASTUTI
Mexico	Mauricio FLORES VILLASUSO
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FAO Resource Experts	Richard ARTHUR Fernando MARDONES Thales PASSOS DE ANDRADE Ramesh PERERA Melba REANTASO Kathy (Feng-Jyu) TANG

Annex 2

Group photos during the Project workshops



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Delegates representing China (Yellow Sea Fisheries Research Institute of the Chinese Academy of Fishery Sciences), Ecuador (National Fisheries Institute), Indonesia (Ministry of Marine Affairs and Fisheries), Mexico (National Health Service, Food Safety and Quality), Thailand (Department of Fisheries), FAO experts from Canada, Chile and the USA and local Brazilians representing the government, producer and academic sectors who participated in a 4-day workshop of the FAO Project "FAO – TCP/INT/3501: Strengthening biosecurity governance and capacities for dealing with the serious shrimp infectious myonecrosis virus (IMNV) disease" held in Brazil in May 2017 in collaboration with MDIC, MAPA and ABCC.



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Country delegates (Brazil, China, Ecuador, Indonesia, Mexico, Thailand), resource experts and Chinese officials (Ms Qingli, Director of Disease Prevention and Control Department of NFTEC; Mr Feng Zhang, Deputy Director of NFTEC and Mr Xianshi Jin, Director of Yellow Sea Fisheries Research Institute) who participated during the final workshop of the FAO Project TCP/INT/3501: Strengthening biosecurity governance and capacities for dealing with the serious shrimp infectious myonecrosis virus (IMNV) disease, held from 2-5 November 2017, Qingdao, China in collaboration with the National Fisheries Technology Extension Center (NFTEC) and the Yellow Sea Fisheries Research Institute (YSFRI).

The Shrimp Infectious Myonecrosis Strategy Manual provides supporting information for the development of contingency plans for countries, producers and other stakeholders with regard to outbreaks of infectious myonecrosis (IMN) in penaeid shrimp. This manual is produced through the Food and Agriculture Organization of the United Nations (FAO) project TCP/INT/3501: “Strengthening biosecurity governance and capacities for dealing with the serious shrimp infectious myonecrosis virus (IMNV) disease”. The document describes current information relevant to reducing the risk of, and economic losses from, outbreaks of IMN. Included are: current knowledge of IMNV and its susceptible host species; detailed diagnostic methods; epidemiology of the IMNV; methods of prevention and management of the disease; planning of national strategies for aquatic animal health (NSAAH) and biosecurity implementation; and discussions of past and potential economic impacts. The manual will also assist countries by providing a framework for the development of national strategies for responses to IMN outbreaks.

ISBN 978-92-5-131798-3 ISSN 2070-6065



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CA6052EN/1/09.19